

Operations Research I

Fundamentals of Simplex: A Worked Problem

IE 311

Consider the following problem:

$$\begin{array}{ll}\max & z = 4x_1 - x_2 + 2x_3 \\ \text{s.t.} & 2x_1 - 2x_2 + 3x_3 \leq 5, \\ & x_1 + x_2 - x_3 \leq 3, \\ & x_1 - x_2 + x_3 \leq 2, \\ & x_1 + 2x_2 - x_3 \geq 1, \\ & x_1, x_2, x_3 \geq 0.\end{array}$$

An Intermediate Tableau

Let x_4 , x_5 , and x_6 denote the slack variables for constraints 1, 2, and 3, respectively. Furthermore, let x_7 denote the surplus variable for constraint 4. After you apply the simplex method, a portion of an intermediate simplex tableau is as follows:

z	x_1	x_2	x_3	x_4	x_5	x_6	x_7	Right Side
1				1	1	0	0	
0		1		1		0	0	
0		0	0		1		0	
0	4		1	1	2			
0				1		0	1	16